Dataset Preparation guidelines for the project LuminaAI

Step 1: - Create a Word Document on your system with the document name as your textbook assigned. Note down the subtopic from where the questions will be extracted as shown below.
Step 2: - Download the scanned images provided to you to your system for text extraction

Step 3: - Open the images and note down the Subtopic that you will be extracting the questions.

In the above image,**Physical world(chapter name) is the topic and what is Physice? Is subtopic**, so whatever questions are extracted until the next subtopic add the extracted questions under the subtopic in the word file you have created.

Step 4: - Go to the website link

Upload the images to extract text, you can upload a maximum of **10 images** at

once.

Make sure u upload the images that has questions along with the respective solutions that is provided on end of the chapter since solutions to those are provided at the end of the chapter

Step 5: - After the image upload is done, click on the convert to text and wait for the output.

Each image's output will be shown as above, copy the output using the above-highlighted button.

Step 6:- Paste the copied text from the extracted image into this document

Step 7:-Copy 5 questions at a time along with respective solutions from this document and paste it to chatgpt

Step 8: - Once the 5 questions and answers are copied use the prompt below and the copied content to extract the questions and solutions in a proper format.

Prompt: For Physics

I have extracted multiple-choice questions (MCQs) and their solutions from the **MTG Fingertips** book for **Physics NEET** preparation. I need you to format the data properly, ensuring that each question is matched with its correct solution. Additionally:

1. **For numerical or problem-based questions**, provide a **step-by-step solution** with clear explanations for each step.
2. **For theoretical questions**, provide a **detailed concept explanation** and explain **why the incorrect options are wrong** to reinforce conceptual clarity and eliminate confusion.
3. **For conceptual problems**, explain the **underlying physics laws and principles** in a way that helps students understand the problem as a whole.
4. **For reaction-based or derivation-based problems**, provide a **structured explanation** that connects fundamental principles with the given question.
5. **Classify** the difficulty level of each question as **Easy, Medium, or Difficult** based on its complexity and relevance to NEET.
6. **Ensure that the final formatted text** copies correctly without breaking in any document or text editor.

For Numerical Questions (Step-by-Step Solution Required)

Difficulty Level: Medium
Question:
A car starts from rest and accelerates uniformly at 5 meters per second squared for 4 seconds. What is its final velocity?
(a) 10 meters per second
(b) 15 meters per second
(c) 20 meters per second
(d) 25 meters per second

Answer: (c)

Step-by-Step Solution:

- **Given:**
  - Initial velocity (u) = 0 meters per second
  - Acceleration (a) = 5 meters per second squared
  - Time (t) = 4 seconds
- **Using the first equation of motion:**
  $v = u + at$
- **Substituting the values:**
  $v = 0 + (5 \times 4)$
  $v = 20$ meters per second

Final Answer: 20 meters per second (c)

For Conceptual Problems (Explain the Whole Concept Clearly)

Difficulty Level: Medium
Question:
A ball is thrown vertically upward with a velocity of 20 meters per second. What will be its velocity at the highest point?
(a) 0 meters per second
(b) 10 meters per second
(c) 20 meters per second
(d) 30 meters per second

Answer: (a)

Concept Explanation:

- When a ball is thrown upwards, it moves against gravity.
- At the **highest point**, the velocity becomes **zero** because the ball momentarily stops before falling back.
- This is a direct result of the **kinematics equation** where final velocity at the highest point is always **zero** for an upward motion under gravity.

Final Answer: 0 meters per second (a)

Data Validation Checklist:

Ensure every question has:

- A **complete and correctly formatted** question text.
- **All answer options** clearly listed.
- The **correct answer** mapped to the options.
- A **step-by-step solution** for numerical questions.
- A **detailed concept explanation** for theoretical and conceptual questions.
- **Why incorrect options are wrong** explained only for theoretical questions.

Ensure clarity, accuracy, and NEET relevance in all answers.

Prompt: For Biology

"Copied Text"

I have extracted multiple-choice questions (MCQs) and their solutions from the **MTG Fingertips** book for **Biology NEET** preparation. I need you to format the data properly, ensuring that each question is matched with its correct solution. Additionally:

1. Provide a comprehensive explanation related to the concept in the question. 2. Include any

missing details that help in understanding the concept thoroughly. 3. Explain why the incorrect options are wrong to eliminate confusion and reinforce conceptual clarity.

4. Classify the difficulty level of each question as Easy, Medium, or Difficult based on its complexity and relevance to NEET.
5. Ensure that the final formatted text copies correctly without breaking in any document or text editor.

Expected Output Format

Difficulty Level: Easy
Question:
Which organelle is known as the powerhouse of the cell?
(a) Nucleus
(b) Ribosome
(c) Mitochondrion
(d) Golgi apparatus

Answer: (c)

Concept Explanation:
The mitochondrion is responsible for aerobic respiration, producing ATP, which is the main energy source for cellular activities.
It has a double membrane structure. The outer membrane is smooth, while the inner membrane has folds called cristae, which increase surface area for ATP synthesis.
The matrix contains enzymes, mitochondrial DNA, and ribosomes, allowing mitochondria to produce some of their own proteins.
Because mitochondria generate energy, they are often referred to as the powerhouse of the cell.

Why Other Options Are Incorrect:
(a) Nucleus: The nucleus contains DNA and controls cellular activities but does not produce energy. It is the control center of the cell, not the powerhouse.
(b) Ribosome: Ribosomes are responsible for protein synthesis, not ATP production. They are not involved in energy generation.
(d) Golgi apparatus: The Golgi apparatus is responsible for modifying, packaging, and transporting proteins and lipids, but it does not generate ATP.

Final Answer: Mitochondrion (c)

Data Validation Checklist:

Ensure every question has:

- A complete and correctly formatted question text.
- All answer options clearly listed.
- The correct answer mapped to the options.
- A detailed concept explanation covering all necessary points.

- An explanation of why the incorrect options are wrong to reinforce learning.

Ensure clarity, accuracy, and NEET relevance in all answers.

Prompt: For Chemistry

I have extracted multiple-choice questions (MCQs) and their solutions from the **MTG Fingertips** book for **Chemistry NEET** preparation. I need you to format the data properly, ensuring that each question is matched with its correct solution. Additionally:

1. **For numerical or problem-based questions**, provide a **step-by-step solution** with clear calculations.
2. **For theoretical questions**, provide a **detailed concept explanation** related to the topic.
3. **For reaction-based questions**, clearly explain the **reaction mechanism, reactants, products, and any important conditions** involved.
4. **Explain why the incorrect options are wrong** to eliminate confusion and reinforce conceptual clarity.
5. **Classify** the difficulty level of each question as **Easy, Medium, or Difficult** based on its complexity and relevance to NEET.
6. **Ensure that the final formatted text** copies correctly without breaking in any document or text editor.

Expected Output Format

Difficulty Level: Medium
Question:
Which product is formed when ethanol is heated with concentrated sulfuric acid at 170°C?
(a) Ethene
(b) Ethanoic acid
(c) Methane
(d) Ethanol

Answer: (a)
Reaction Explanation:

- When ethanol is heated with **concentrated H_2SO_4 at 170°C**, it undergoes **dehydration** to form **ethene (C_2H_4)**and water.
- **Reaction:**
  $C_2H_5OH \rightarrow C_2H_4 + H_2O$ (in presence of conc. H_2SO_4 and heat)
- **Mechanism:**
  - Sulfuric acid acts as a **dehydrating agent**, removing water.
  - The **hydroxyl group (-OH)** and a hydrogen atom from adjacent carbon atoms are eliminated, forming a **double bond**.

Why Other Options Are Incorrect:
(b) Ethanoic acid: Ethanol gets oxidized to ethanoic acid in the presence of an oxidizing agent like potassium dichromate, not sulfuric acid alone.
(c) Methane: There is no reaction mechanism in which ethanol converts to methane under these conditions.
(d) Ethanol: Ethanol is the reactant, not the product in this reaction.

Final Answer: Ethene (a)

Data Validation Checklist:

Ensure every question has:

- A **complete and correctly formatted** question text.
- **All answer options** clearly listed.
- The **correct answer** mapped to the options.
- A **step-by-step solution** for numerical questions.
- A **detailed concept explanation** for theoretical questions.
- A **reaction explanation** for reaction-based questions.
- An explanation of **why the incorrect options are wrong** to reinforcelearning.

Ensure clarity, accuracy, and NEET relevance in all answers.

Step 9: Make sure chatgpt generates text in the following Format:

Difficulty Level: Difficult

Question:

Two stable isotopes ^{6}Li and ^{7}Li have respective abundances of 7.5% and 92.5%. These isotopes have masses 6.01512 u and 7.01600 u respectively. The atomic weight of lithium is:

(a) 6.941 u
(b) 3.321 u
(c) 2.561 u
(d) 0.621 u

Answer:

(a)

Solution:

1. Atomic weight is calculated using the formula:

$$\text{Atomic weight} = (m_1 \times f_1) + (m_2 \times f_2)$$

Where:

- $m_1 = 6.01512\,u$
- $f_1 = 0.075$
- $m_2 = 7.01600\,u$
- $f_2 = 0.925$

2. Substituting the values:

$$\text{Atomic weight} = (6.01512 \times 0.075) + (7.01600 \times 0.925)$$

3. Performing the calculations:

$$\text{Atomic weight} = (0.451134) + (6.4898)$$

$$\text{Atomic weight} = 6.940934 \approx 6.941\,u$$

Final Answer: The atomic weight of lithium is approximately 6.941 u.

Step 10:- Copy the output from GPT and paste it to the Excel sheet provided correctly under each field **Topic**-Chapter name,**Subtopic,Question/Prompt** along with the options,**Correct answer,Step by step solution,Difficulty Level**. Make sure that you put the questions and solutions in a proper format ,follow the same formatting till the end.

Step 11:-If there are any mathematical expressions,powers,subscripts,superscripts,..... Etc. that cant be pasted directly apply latex formatting to it.

Step 12: - Repeat the process for all the images.

End of Prep.